

Identify and describe the properties of prime, composite and square numbers and use these properties to solve problems and simplify calculations

Learning intention

- identify and describe the properties of prime, composite and square numbers and use these properties to solve problems and simplify calculations.

Teach and model

- representing composite numbers as a product of their factors.
- identifying and describing the product of a number with itself as square.
- using the definition of a prime number to explain why one is not a prime number.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.